

Pearson Edexcel International Advanced Level

Wednesday 22 October 2025

Afternoon (Time: 1 hour 45 minutes)

Paper
reference

WCH14/01A

Chemistry

International Advanced Level

UNIT 4: Rates, Equilibria and Further Organic Chemistry

Question Paper

You must have:

Scientific calculator, Data Booklet, ruler

Answer Booklet (sent separately)

Correction to Answer Book

Question 19e is worth 3 marks **NOT** 2 marks.

Question 19 and Section C are worth a total of 21 marks.

Turn over ►

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M:1/1/1/



P 8 7 4 2 1 A


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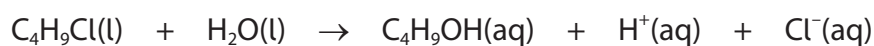
SECTION A

Answer ALL the questions in this section.

You should aim to spend no more than 20 minutes on this section.

For each question, select one answer from A to D and put a cross in the box ☐. If you change your mind, put a line through the box ☒ and then mark your new answer with a cross ☐.

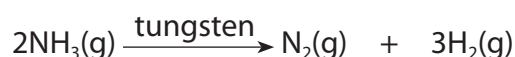
- 1 Which **continuous** monitoring method could be used to obtain rate data for the reaction shown?



- A colorimetry
- B electrical conductivity
- C mass change
- D titration with alkali

(Total for Question 1 = 1 mark)

- 2 The decomposition of ammonia is catalysed by tungsten metal.



This reaction has zero order kinetics.

- (a) What is the rate equation for this reaction?

(1)

- A rate = k
- B rate = $k[\text{NH}_3]$
- C rate = $k[\text{NH}_3]^2$
- D rate = $k[\text{N}_2][\text{H}_2]^3$

- (b) What are the units of the rate constant, k , for this zero order reaction?

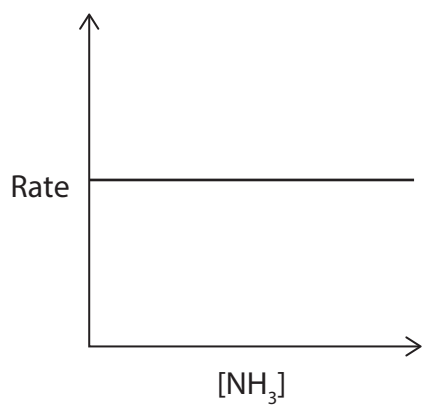
(1)

- A no units
- B s^{-1}
- C $\text{mol dm}^{-3} \text{s}^{-1}$
- D $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$

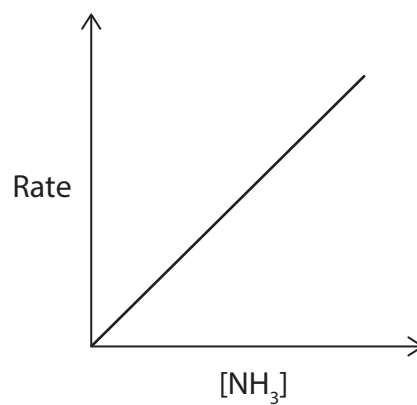


(c) Which of these graphs represents this zero order reaction?

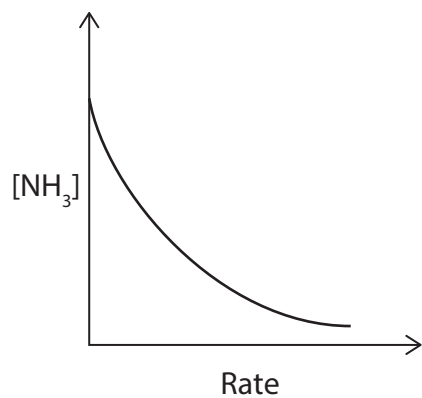
(1)



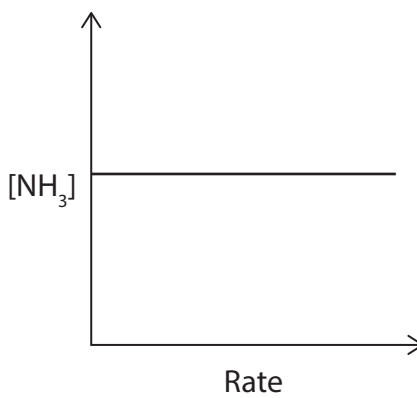
A



B



C



D

(Total for Question 2 = 3 marks)

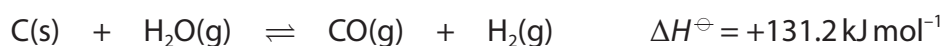
- 3 The halogenoalkane 2-bromo-2-methylbutane was hydrolysed with sodium hydroxide solution, NaOH(aq).

Which suggestion about the mechanism of this reaction is correct?

	Type of mechanism	Number of steps in mechanism
A	S _N 2	one
B	S _N 2	two
C	S _N 1	one
D	S _N 1	two

(Total for Question 3 = 1 mark)

- 4 The water gas reaction is used in the manufacture of hydrogen.

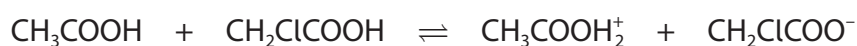


What is the equilibrium constant, K_c , expression for this reaction?

- A $K_c = [\text{CO}][\text{H}_2]$ B $K_c = \frac{[\text{CO}][\text{H}_2]}{[\text{C}]}$
- C $K_c = \frac{[\text{CO}][\text{H}_2]}{[\text{H}_2\text{O}]}$ D $K_c = \frac{[\text{CO}][\text{H}_2]}{[\text{H}_2\text{O}][\text{C}]}$

(Total for Question 4 = 1 mark)

- 5 When ethanoic acid and chloroethanoic acid are mixed, an equilibrium is set up.



The Brønsted-Lowry acids in this equilibrium are

- A CH₃COOH and CH₂ClCOOH
- B CH₃COOH and CH₃COOH₂⁺
- C CH₂ClCOOH and CH₃COOH₂⁺
- D CH₃COOH₂⁺ and CH₂ClCOO⁻

(Total for Question 5 = 1 mark)



6 What is the pH of a 0.05 mol dm^{-3} solution of barium hydroxide?

- A 13.3
- B 13.0
- C 12.7
- D 12.4

(Total for Question 6 = 1 mark)

7 What is the pH of a solution containing $0.100 \text{ mol dm}^{-3}$ CH_3COOH and $0.200 \text{ mol dm}^{-3}$ CH_3COONa ?

[Data: $\text{p}K_{\text{a}} \text{CH}_3\text{COOH} = 4.76$]

- A 5.06
- B 4.76
- C 4.46
- D 2.88

(Total for Question 7 = 1 mark)

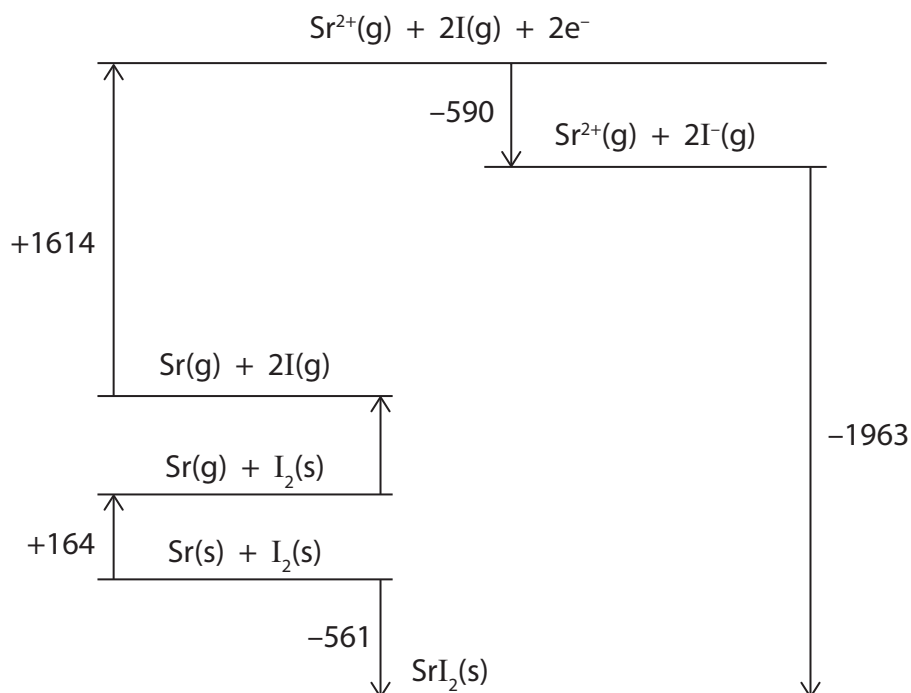
8 Which of these indicators is **most** suitable for the titration of ammonia solution with hydrochloric acid?

- A methyl violet ($\text{p}K_{\text{in}} = 0.8$)
- B bromocresol green ($\text{p}K_{\text{in}} = 4.7$)
- C thymol blue ($\text{p}K_{\text{in}} = 8.9$)
- D alizarin yellow ($\text{p}K_{\text{in}} = 12.5$)

(Total for Question 8 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

- 9 The Born-Haber cycle for strontium iodide, SrI_2 , is shown. All values are in kJ mol^{-1} . The cycle is not to scale.



- (a) Which of the following statements is **not** true?

(1)

- A** the lattice energy of strontium iodide is exothermic
- B** the first electron affinity of iodine is -295 kJ mol^{-1}
- C** the second ionisation energy of strontium is $+1614 \text{ kJ mol}^{-1}$
- D** the standard enthalpy change of atomisation of strontium is $+164 \text{ kJ mol}^{-1}$

- (b) Using the Born-Haber cycle, it can be calculated that the standard atomisation of iodine, in kJ mol^{-1} , is

(1)

- A** $+542$
- B** $+271$
- C** $+214$
- D** $+107$

(Total for Question 9 = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.

10 This question is about chromatography.

- (a) In gas chromatography, the time taken for the substance to travel from the input to the detector is called the

(1)

- A** reference time
- B** retention time
- C** separation time
- D** travelling time

- (b) Which of the following would be the most suitable carrier gas in gas chromatography?

(1)

- A** argon
- B** hydrogen
- C** methane
- D** oxygen

- (c) A spot caused by an amino acid has moved 52 mm from the baseline of a paper chromatogram.

The R_f value for the amino acid under these conditions is 0.62.

What is the distance moved by the solvent?

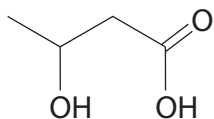
(1)

- A** 840 mm
- B** 84 mm
- C** 52 mm
- D** 32 mm

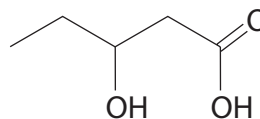
(Total for Question 10 = 3 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.

- 11** The substance known as PHBV is a biodegradable polymer formed from 3-hydroxybutanoic acid and 3-hydroxypentanoic acid.

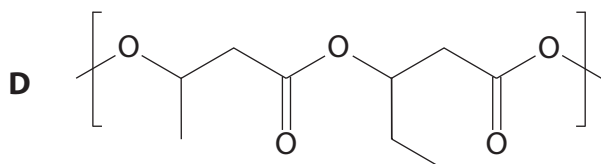
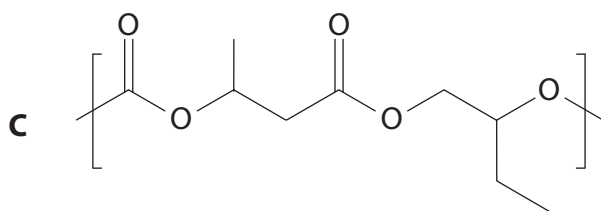
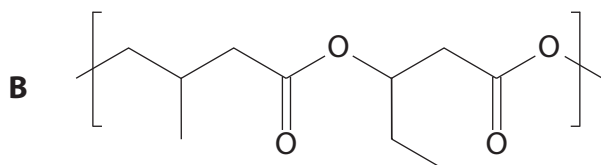
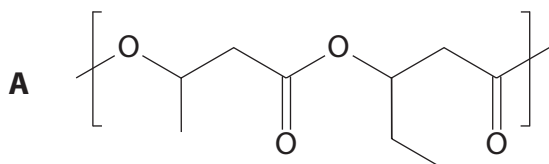


3-hydroxybutanoic acid



3-hydroxypentanoic acid

Which of these is the repeat unit of the polymer?



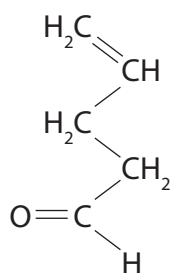
(Total for Question 11 = 1 mark)

- 12** Which reagent reacts at room temperature with methylamine, CH_3NH_2 , to form the compound N-methylethanamide?

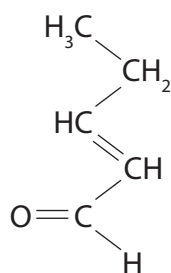
- A** CH_3COCH_3
B CH_3COOH
C $\text{CH}_3\text{COOCH}_3$
D CH_3COCl

(Total for Question 12 = 1 mark)

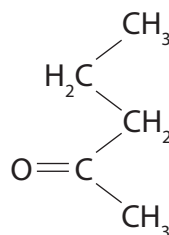
13 This question refers to the following molecules, all of which have a carbonyl group.



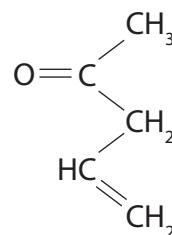
W



X



Y



Z

(a) A pale yellow precipitate is formed when iodine in alkali reacts with

(1)

- A** **W** and **X** only
- B** **Y** only
- C** **Y** and **Z** only
- D** **Z** only

(b) A brick red precipitate is formed on warming blue Benedict's or Fehling's solution with

(1)

- A** **W** only
- B** **X** only
- C** **W** and **X** only
- D** **Y** and **Z** only

(c) A mass spectrum fragment with a m/z value of 29 would be expected for

(1)

- A** **W** and **X** only
- B** **X** and **Y** only
- C** **Y** and **Z** only
- D** **W**, **X** and **Y** only

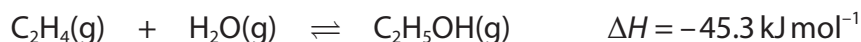
(Total for Question 13 = 3 marks)

TOTAL FOR SECTION A = 20 MARKS

SECTION B

Answer ALL the questions. Write your answers in the spaces provided.

14 Ethanol may be produced by the hydration of ethene.



- (a) In a laboratory investigation of this reaction, 1.00 mol of ethene was mixed with 1.00 mol of steam at 160 °C. At equilibrium, when the total pressure of the steam was 50.0 atm, 0.430 mol of ethanol had formed.

(i) Give the expression for the equilibrium constant, K_p , for the reaction.

(1)

- (ii) Calculate the equilibrium constant, K_p , for the hydration of ethene at 160 °C. Include units in your answer.
You **must** show your working.

(5)

- (b) The manufacture of ethanol is carried out at 230 °C and 70 atm, with an overall conversion into ethanol of 95%.

Comment on these conditions in relation to their effect on the equilibrium and the overall yield of ethanol.

(3)

(Total for Question 14 = 9 marks)

15 Butanedione, $\text{CH}_3\text{COCOCH}_3$, has two carbonyl groups. It is a volatile yellow-green liquid and its colour is due to electron delocalisation.

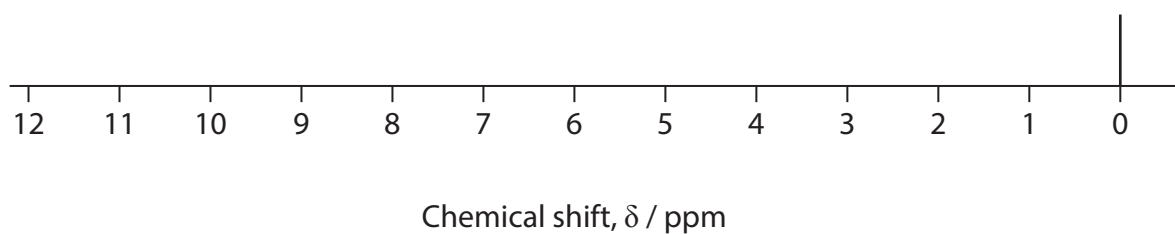
Butanedione can be reduced to butane-2,3-diol, which does not have this electron delocalisation.

- (a) Complete the equation in the Answer Book for this reduction of butanedione. (2)
- (b) Suggest what you would see when this reaction occurs. (1)
- (c) A mixture of butanedione and butane-2,3-diol can be separated by distillation.
Explain which compound has the higher boiling temperature. (2)
- (d) Butane-2,3-diol shows a type of stereoisomerism that butanedione does not.
Explain this type of stereoisomerism shown. (2)
- (e) Butane-2,3-diol can be esterified using propanoyl chloride or propanoic acid.
- (i) Identify three differences in the esterification reaction when propanoyl chloride is used instead of propanoic acid. (3)
- (ii) Draw the **skeletal** formula of the ester produced when butane-2,3-diol reacts with **excess** propanoic acid. (2)
- (f) Give **two** differences between the infrared spectra of butanedione and butane-2,3-diol. Include wavenumbers and the bonds responsible.
Use your Data Booklet. (2)

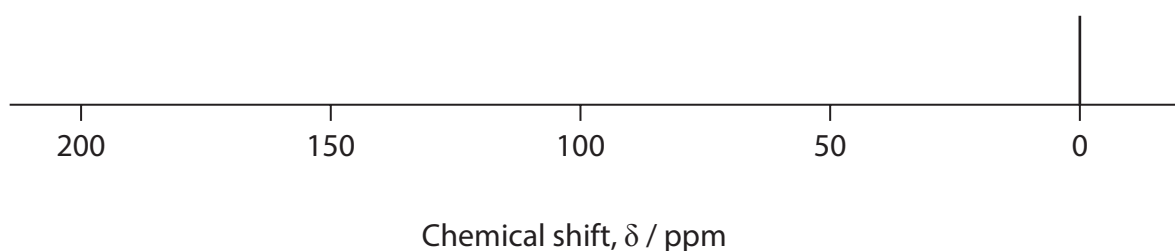
- (g) Complete both the high resolution ^1H spectrum **and** the ^{13}C NMR spectrum for butanedione. Include the relevant hydrogen atoms and carbon atoms responsible for the peaks. Use your Data Booklet.

(3)

^1H spectrum



^{13}C spectrum



(Total for Question 15 = 17 marks)

- *16** The table shows the theoretical and experimental (Born-Haber) lattice energy data for two metal halide compounds, potassium chloride and calcium iodide.

Metal halide	Lattice energy / kJ mol^{-1}	
	Theoretical	Experimental (Born-Haber)
Potassium chloride	-702	-711
Calcium iodide	-1905	-2074

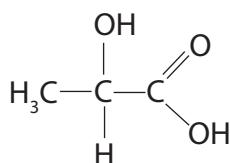
Using the data, compare and contrast the type and strength of bonding in these compounds.

Include reasons for your answers.

(6)

(Total for Question 16 = 6 marks)

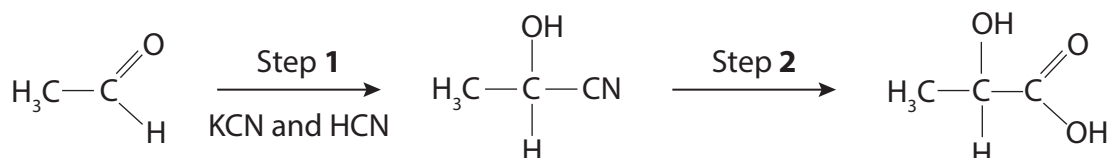
17 Lactic acid has the structure shown.



(a) Give the IUPAC name of lactic acid.

(1)

(b) Lactic acid can be synthesised from ethanal in two steps.



(i) Draw the mechanism for Step 1.
Include curly arrows and relevant lone pairs and dipoles.

(4)

(ii) State the reaction type for Step 2.

(1)

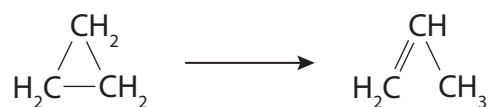
(c) Sodium hydrogencarbonate, NaHCO_3 , has been used to prevent lactic acid causing muscle pain during exercise.

Write an equation for the reaction between sodium hydrogencarbonate and lactic acid.

(1)

(Total for Question 17 = 7 marks)

- 18** The rate constant for the reaction to convert cyclopropane to propene was determined at five different temperatures.



The results are shown in the table.

Temperature, T / K	$1 \div T$ / K ⁻¹	Rate constant, k / s ⁻¹	ln k
714	1.40×10^{-3}	1.67×10^{-5}	-11.00
746	1.34×10^{-3}	1.23×10^{-4}	-9.00
791		1.66×10^{-3}	-6.40
833	1.20×10^{-3}	1.23×10^{-2}	
893	1.12×10^{-3}	1.65×10^{-1}	-1.80

- (a) (i) Complete the table. (1)
- (ii) Plot a graph of ln k against $1 \div T$. (3)
- (iii) Determine the activation energy, E_a , in kJ mol⁻¹ using your graph.
You should include the value and units of the gradient of the line.
The Arrhenius equation is

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant}$$
(3)
- (b) Explain the trend in the value of the rate constant k as the temperature increases. (3)

(Total for Question 18 = 10 marks)

TOTAL FOR SECTION B = 49 MARKS

SECTION C

Answer ALL the questions. Write your answers in the spaces provided.

19 Endothermic changes can be thermodynamically feasible.

- (a) One example of an endothermic reaction is between the solids hydrated barium hydroxide and ammonium chloride.

The equation for this reaction is shown.



- (i) Suggest **two** experimental observations that you would expect to make when carrying out this reaction without additional chemical tests.

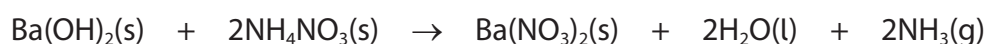
(2)

- (ii) Predict the sign of the standard entropy change of the system ($\Delta S_{\text{system}}^\ominus$) for this reaction. Justify your choice with **two** reasons.

(2)

- (b) Another example of a reaction between two solids involves anhydrous barium hydroxide and ammonium nitrate.

The equation for this reaction is shown.



Compound	Standard molar entropy, $S^\ominus$ / $\text{J K}^{-1} \text{mol}^{-1}$	Standard enthalpy change of formation, $\Delta_f H^\ominus$ / kJ mol^{-1}
$\text{Ba(OH)}_2\text{(s)}$	99.7	−944.7
$\text{NH}_4\text{NO}_3\text{(s)}$	151.1	−365.6
$\text{Ba(NO}_3)_2\text{(s)}$	213.8	−992.1
$\text{H}_2\text{O(l)}$	69.9	−285.8
$\text{NH}_3\text{(g)}$	192.3	−46.1

- (i) Calculate the standard molar entropy change of the system, $\Delta S_{\text{system}}^\ominus$, for this reaction at 298 K.

Use the data in the table.

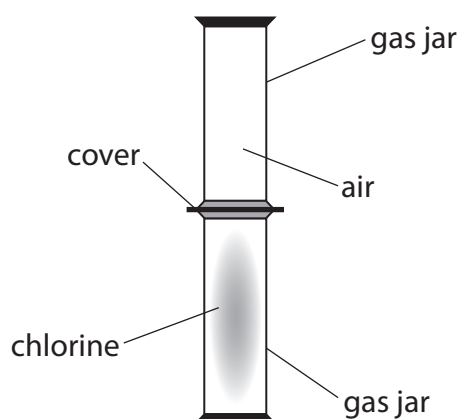
(3)

(ii) Calculate the standard molar enthalpy change of this reaction at 298 K. (2)

(iii) Calculate the total entropy change for this reaction, $\Delta S_{\text{total}}^{\ominus}$, at 298 K using your answers to (b)(i) and (b)(ii).
Include a sign and units in your answer. (3)

(c) The feasibility of chemical changes can be related to entropy in terms of the dispersal of molecules and of energy quanta between molecules.

(i) Explain, in terms of entropy, what happens to the chlorine gas when the cover between the gas jars is removed. (2)



(ii) Complete the table to show the five different ways that **four** energy quanta can be shared between two molecules. (1)

Molecule A	Molecule B
2	2
3	1

- (d) Draw a sketch of entropy against temperature for water to illustrate the changes in entropy as the temperature changes.

On your sketch indicate where water changes state.

A scale is not required for the vertical axis.

(3)

- (e) Explain why the entropy change of the system increases when sodium chloride dissolves.

(3)

(Total for Question 19 = 21 marks)

TOTAL FOR SECTION C = 21 MARKS

TOTAL FOR PAPER = 90 MARKS



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The Periodic Table of Elements

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Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International Advanced Level

Wednesday 22 October 2025

Afternoon (Time: 1 hour 45 minutes) **Paper reference** **WCH14/01A**

Chemistry
International Advanced Level
UNIT 4: Rates, Equilibria and Further Organic Chemistry
Answer Book

You must have:
 Question paper (sent separately), Scientific calculator,
 Data Booklet, ruler

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
 – *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
 – *use this as a guide as to how much time to spend on each question.*
- In the question marked with an **asterisk** (*), marks will be awarded for your ability to structure your answer logically, showing how the points that you make are related or follow on from each other where appropriate.
- A Periodic Table is printed on the back cover of this paper.

Advice

- Read each question carefully before you start to answer it.
- Show all your working in calculations and include units where appropriate.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION A

Answer ALL the questions in this section.

You should aim to spend no more than 20 minutes on this section.

For each question, select one answer from A to D and put a cross in the box ☐. If you change your mind, put a line through the box ☒ and then mark your new answer with a cross ☐.

1

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 1 = 1 mark)

2

(a)

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(b)

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D



(c)

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 2 = 3 marks)

3

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 3 = 1 mark)

4

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 4 = 1 mark)

5

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 5 = 1 mark)



6

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 6 = 1 mark)

7

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 7 = 1 mark)

8

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 8 = 1 mark)

9

(a)

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D



(b)

(1)

- ☐ A
☐ B
☐ C
☐ D

(Total for Question 9 = 2 marks)

10

(a)

(1)

- ☐ A
☐ B
☐ C
☐ D

(b)

(1)

- ☐ A
☐ B
☐ C
☐ D

(c)

(1)

- ☐ A
☐ B
☐ C
☐ D

(Total for Question 10 = 3 marks)



11

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 11 = 1 mark)

12

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 12 = 1 mark)

13

(a)

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(b)

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D



(c)

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 13 = 3 marks)

TOTAL FOR SECTION A = 20 MARKS

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



SECTION B

Answer ALL the questions. Write your answers in the spaces provided.

14

(a)

(i)

(1)

(ii)

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b)

(3)

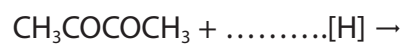
(Total for Question 14 = 9 marks)



15

(a)

(2)



(b)

(1)

(c)

(2)

(d)

(2)



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(e)

(i)

(3)

(ii)

(2)

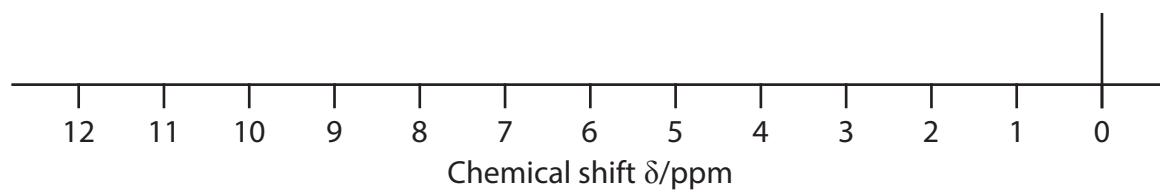
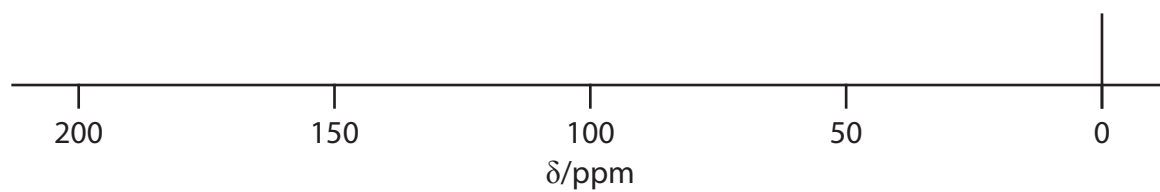


(f)

(2)

(g)

(3)

 ^1H spectrum ^{13}C spectrum**(Total for Question 15 = 17 marks)**

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*16

(6)

Handwriting practice area with 20 horizontal dotted lines.



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(Total for Question 16 = 6 marks)



17

(a)

(1)

(b)

(i)

(4)

(ii)

(1)

(c)

(1)

(Total for Question 17 = 7 marks)

18

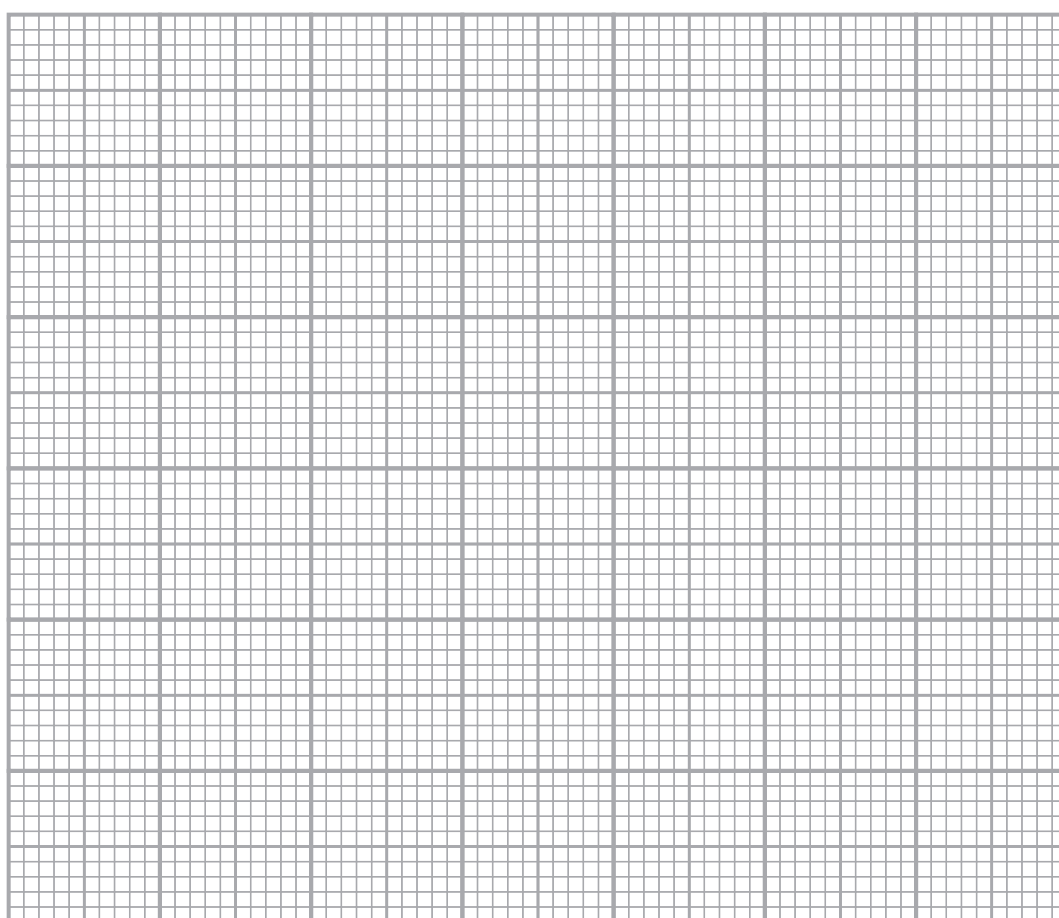
(a)

(i)

(1)

(ii)

(3)



(iii)

(3)

(b)

(3)

(Total for Question 18 = 10 marks)

TOTAL FOR SECTION B = 49 MARKS

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SECTION C

Answer ALL the questions. Write your answers in the spaces provided.

19

(a)

(i)

(2)

(ii)

(2)

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(b)

(i)

(3)

(ii)

(2)



(iii)

(3)

(c)

(i)

(2)

(ii)

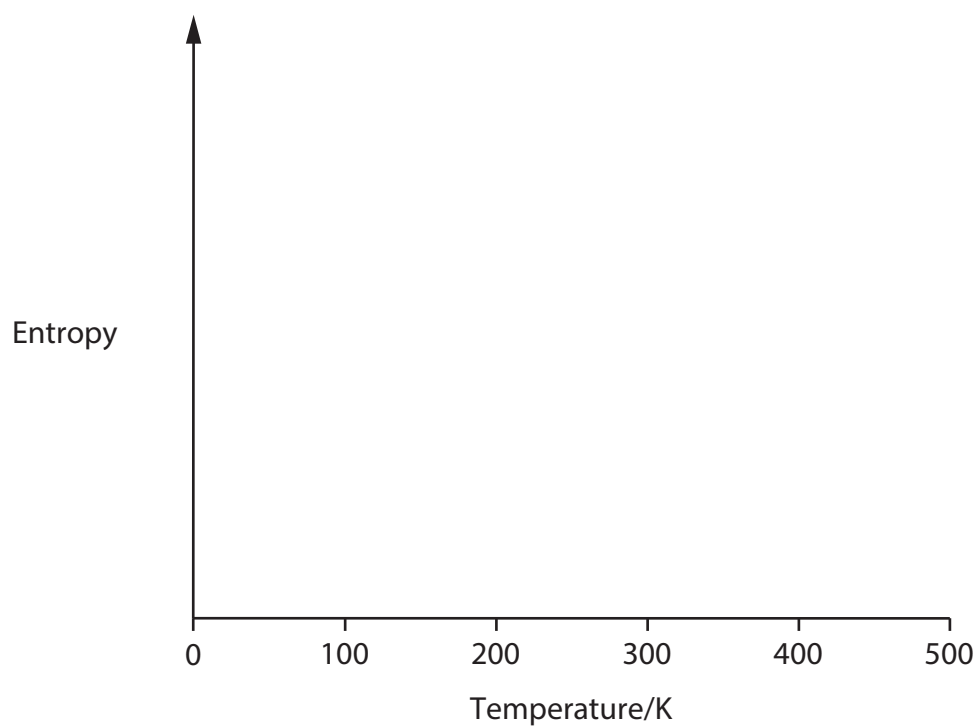
(1)

2	2
3	1



(d)

(3)



(e)

(2)

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(Total for Question 19 = 21 marks)

TOTAL FOR SECTION C = 20 MARKS
TOTAL FOR PAPER = 90 MARKS



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The Periodic Table of Elements

[illegible]

Elements with atomic numbers 112-116 have been reported but not fully authenticated

140	141	144	[147]	150	152	157	159	163	165	167	169	173	175
Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
cerium	praseodymium	neodymium	promethium	samarium	euroium	gadolinium	terbium	dysprosium	holmium	erbium	thulium	ytterbium	lutetium
58	59	60	61	62	63	64	65	66	67	68	69	70	71

232	[231]	238	[237]	[242]	[243]	[247]	[245]	[251]	[254]	[253]	[256]	[254]	[257]
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
thorium	protactinium	uranium	neptunium	plutonium	americium	curium	berkelium	californium	einsteinium	fermium	mendeleevium	nobelium	lawrencium
90	91	92	93	94	95	96	97	98	99	100	101	102	103

* Lanthanide series

* Actinide series

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